Pose coverage and held-out image validation for multi-camera robot calibration

Purpose To test whether pose-coverage effects in existing multi-camera robot calibration records persist across additional hardware archives and validation endpoints.

Design/methodology/approach A fixed retrospective protocol was extended from 12 to 36 camera-session panels across eight robot archives. All 6,383 accepted images were reconstructed. Comparisons used 990 training and 288 held poses, five full-set solvers, 14,832 subset fits and training-only pixel refinement. Another 23 partial-session panels tested sensitivity.

Findings At ten poses, farthest-point selection gave lower translation closure than the random median in 9/12 original and 17/24 added panels. Held-image error favoured it in 17/24 added panels. Added-cohort median ratios were 0.879 for both endpoints. Full-set pixel refinement gave 0.247–0.616 pixel RMS. A paired ten-pose check shifted the median image-error ratio from 0.911 under Park to 0.993 after pixel refinement.

Research limitations/implications Historical holdouts were local and previously inspected. Archive extension does not supply external metrology or simultaneous inter-camera validation.

Originality/value The evaluation links raw-image evidence to pose-selection claims and shows their dependence on fitting objective and validation conditions.

Keywords Hand-eye calibration; Multi-camera robots; Pose coverage; Held-out validation; Reprojection error; Reproducibility

Article classification Research paper

1 Introduction

Cameras mounted on robot heads, torsos and wrists require consistent relationships to supporting links for perception and manipulation. Saved calibration records often contain many images, a fitted transform and a small residual. Interpreting those residuals requires a clear account of pose diversity, validation geometry and image processing.

Industrial calibration studies connect estimated geometry to physical measurements or application performance. Bai *et al.* (2020) combined calibration and detection for mobile robotic manufacturing. Wang *et al.* (2025) evaluated robot self-calibration using a line-structured light sensor. Such application-level evidence provides a useful distinction for archived datasets: internal transform consistency can be recomputed without establishing comparable end-effector positioning accuracy.

Pose selection has a substantial methodological history. Wu *et al.* (2015) examined measurement configurations for robot geometric calibration, while Yang *et al.* (2023) selected informative eye-in-hand views through predicted uncertainty reduction. Li *et al.* (2026) investigated next-best-pose calibration using reinforcement learning. Existing production records present a different problem: their trajectories are fixed, external ground truth may be absent, and historical validation results may already have been inspected. An apparent sampling advantage should therefore be checked on additional records and against more than one endpoint.

This study asks three questions. Does the small-budget pose-coverage pattern observed in an initial archive persist in additional robot archives? Does the comparison agree between held-pose closure and original image corners? How do objective choice and transfer between recording dates change the assessment? We preserve the historical splits and fit every model using training observations only. No robot motion or acquisition was initiated for this analysis.

The contribution is a reproducible archive-extension evaluation across four camera roles, nine complete sessions and eight robot archives, with 23 ancillary camera sequences analysed separately. It combines acquisition deduplication, complete accepted-image reconstruction, fixed-budget sampling, explicit failure

accounting and held-image validation. These are individual camera-role calibrations within multi-camera platforms; the study does not introduce a solver or estimate a synchronized camera network.

2 Related work

Classical hand-eye calibration estimates a rigid transformation from paired robot and camera motions. Tsai and Lenz (1989) separated rotation and translation estimation. Park and Martin (1994) formulated the problem on the Euclidean group. Horaud and Dornaika (1995) examined alternative formulations, and Daniilidis (1999) used dual quaternions to couple the rigid-motion components. These methods remain useful reference implementations, but their numerical behaviour depends on the available motions and their implementation.

Enebuse *et al.* (2022) examined how noise and robot motion affect hand-eye calibration accuracy. Their work motivates reporting rotation and translation separately and resisting a universal ranking across datasets. Tabb and Ahmad Yousef (2017) compared iterative robot-world and hand-eye formulations using several evaluation metrics. Their analysis also makes the choice of objective and evaluation metric central to interpretation. We therefore evaluate both fixed-target pose closure and held-image pixel reprojection, estimating the target reference only from training observations.

Active view selection and retrospective subset selection answer different operational questions. Yang *et al.* (2023) predicted views that reduce calibration uncertainty, while Li *et al.* (2026) combined clustering and reinforcement learning for viewpoint and distance selection. Our farthest-point baseline chooses only among existing poses and uses no image-noise model. It is a geometric comparator, rather than a reproduction of either active method. Concentrated subsets stress restricted coverage, and chronological prefixes test dependence on acquisition order.

A multi-camera robot also differs from a jointly calibrated camera network. Allegro *et al.* (2024) integrated shared target-to-end-effector and camera-to-camera constraints for industrial workcells. The present records contain separately collected camera-role sequences with different supporting links. We evaluate their individual calibrations under a common protocol. Shared-platform deployment motivates the study, but no synchronized inter-camera target observations or independently measured relative camera transforms support a network-accuracy claim.

Joint kinematic and extrinsic calibration can use additional patterns to constrain robot parameters as well as sensor transforms (Vieira *et al.*, 2026). Here the recorded kinematics and intrinsics remain fixed. This reduces the fitted parameter set but leaves their systematic errors inside both evaluation endpoints. Published reprojection errors from different sensors, targets and fitted parameter sets are therefore contextual evidence, not numerical benchmarks for the present data.

Reproducibility also depends on the target definition and corner convention. Kalibr documents tag size and relative spacing as target configuration quantities and recommends checking printed dimensions and flatness (ETH Zurich ASL, n.d.). A family name alone does not specify the complete detector configuration. We assess exact image lineage, raw robot transforms, recorded intrinsics and reconstructed image corners before comparing fitted models.

3 Materials and methods

The study was retrospective and was not preregistered. The initial two-robot analysis had already inspected its historical validation outcomes. Before inspecting numerical outcomes in the additional archives, a timestamped machine-readable extension protocol fixed eligibility, deduplication, detector settings, pose budgets, subset policies, fitting and endpoints. The added archives test transfer of that protocol, but their stored validation results had been available to the original calibration workflow. They are not an untouched prospective test. No subset policy was deployed to a robot.

Further processing details and panel-level outcomes are provided in Supplementary_material.docx.

3.1 Records and provenance

The initial cohort comprised Robot-A sessions A1 and A2, recorded on 10 and 20 September 2026, and Robot-B session B1 on 16 October 2026. Six additional archives supplied C1 (10 September), D1 (1 September), E1 (8 September), F1 and G1 (21 September), and H1 (17 October). Each complete session contained head, body and both wrist sequences. The main cohort therefore has 36 camera-session panels from nine sessions. The platforms have two seven-joint arms, a four-joint trunk and a two-joint head. Roles and dates are nested within robot archives, rather than independent hardware replicates.

Discovery initially surveyed an online-device snapshot with 62 entries, including auxiliary computers and duplicate hostnames. Data-root traversal succeeded for 49 entries; 13 remained inaccessible because of authentication or network restrictions. The private evidence package records every attempt, directory scope, permission error and excluded static software tree. Across the accessible resources, 87 retrieved export versions were linked by common raw-recording identities or overlapping image timestamps into 63 capture groups. Four groups lacked localizable raw recordings. Complete four-role sessions supplied the main cohort; 23 eligible partial-session panels supplied an ancillary analysis. Numerical residuals and workflow PASS labels were not eligibility criteria. Availability of an observation export nevertheless introduces survivorship bias.

The main recordings contain 240,118 image messages. Historical capture windows selected 6,390 candidates at 1,278 poses and accepted 6,383 frames: 4,944 training and 1,439 validation frames. Seven rejected candidates remain in the inclusion log. Each head and body sequence has 25 training poses, each wrist sequence has 30, and every panel has eight validation poses. This is a quality-screened capture-window cohort; it does not estimate all-frame detection reliability. The ancillary cohort contains 3,811 accepted frames from additional or incomplete captures, including raw recordings referenced by software regression exports.

Supporting records include timestamps, pose identifiers, annotations, camera matrices, robot-description hashes and nominal target configurations. The colour images are 640 × 480 pixels with a five-coefficient plumb-bob model. The target is a seven-column, five-row grid with 35 mm tags, spacing ratio 0.3 and a legacy tag36h11 payload with two black border cells. Dimensions and flatness were not independently measured. Seven main archive identities have a session-stored robot identifier. D1 has live-capture flags and distinct recorded sensor metadata, but its installation record contains an error instead of a robot identifier. Its exclusion is reported as a provenance sensitivity. The main cohort contained 4,963 JPEG and 1,420 PNG images. Raw payloads were not transcoded; historical .jpg filenames do not define their encoding.

Read-only retrieval recovered each accepted compressed image at its camera-header timestamp, alongside TF, CameraInfo and robot-description records. Detection, calibration and statistical analysis ran locally. Available nonzero chunk CRCs were checked; whole recordings were not copied or hashed end to end. Storage-host identity establishes archive location rather than every historical acquisition condition. Retrieval and integrity details are retained in the evidence ledger and supplement.

The retained detector snapshots recovered in the initial study have matching SHA-256 hashes, and their target-generator codebook is preserved. They define a fixed compatible reprocessing implementation, rather than a verified source commit for every recording date. Raw CameraInfo, robot-description hashes and frame-specific support transforms were checked separately against each export. Failed initialization attempts, partial sessions and copied reference data remain in the acquisition catalogue. Stored workflow status is distinct from manual or metrological acceptance. Plate 1 illustrates one retained wrist-camera view and its reconstructed corner overlay.

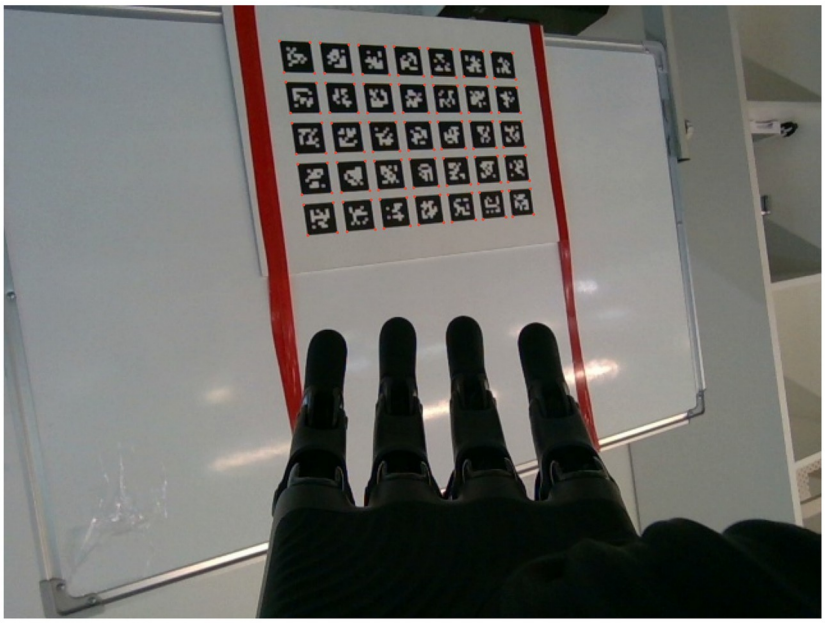


Plate 1. An existing E1 right-wrist camera image with reconstructed target corners in red. The image shows the physical capture and image-processing output; it does not provide independent mounting ground truth. Source: authors' existing recordings and present corner reconstruction.

3.2 Transform model and pose aggregation

Each rigid transform maps coordinates from its child frame to its parent frame. For observation i, G_i maps the camera-support link into the robot base, while C_i maps the target into the camera. X relates camera coordinates to that support link, and Z relates target coordinates to the robot base. The observation model is:

$$G_i X C_i = Z. \qquad (1)$$

The exported robot transform uses the legacy key T_base_Head2 for every camera role. We separately reconstructed base_link-to-support transforms from the copied raw TF messages at each accepted image timestamp, using the recovered nearest-sample lookup with a 200 ms maximum skew. The support frames were Head2, cam_install_body, base_L_wrist and base_R_wrist. The camera coordinate convention is the pinhole optical convention used by PnP and projection, even where a recorded frame identifier has a legacy link suffix. Each raw transform was compared with the exported matrix. Raw CameraInfo fields and robot-description hashes were checked against their exports for every bag.

Frames were grouped by split, run identifier and pose identifier. Translations were averaged arithmetically and rotations by the SciPy chordal mean, separately for robot and target transforms. Each resulting pose received equal weight. Repeated frames thus reduce within-pose variation without increasing the pose count. All-frame weighting and first, middle and last frame checks from the initial cohort are retained as processing sensitivities; the archive extension additionally repeats training-pose deletion.

Eliminating Z from Equation (1) gives relative motions $A_{ij} = G_j^{-1}G_i$ and $B_{ij} = C_jC_i^{-1}$ satisfying $A_{ij}X = XB_{ij}$. The primary estimate used OpenCV 4.11.0 calibrateHandEye with the Park option (OpenCV, 2025a). Input poses retained their recorded order. The target reference was then the mean of G_iXC_i over training poses only. Neither X nor Z was re-estimated using validation observations.

3.3 Coverage descriptors and subset policies

Coverage was described by the maximum pairwise robot rotation angle, maximum pairwise support-link translation distance and distribution of rotation axes. For the latter, a matrix contained the rotation vectors of all relative robot rotations. Its smallest-to-largest singular-value ratio described directional spread. This ratio is a geometric descriptor, not a complete identifiability score. We also recorded the condition number of the stacked translation system formed by relative rotation matrices minus identity.

Training-pose budgets were six, ten, 15 and 20. The concentrated policy selected the k poses closest to the training medoid. The chronological policy selected the first k recorded poses. Random selection sampled k poses without replacement, using 100 subsets for every panel and budget. Farthest-point selection began with the pose farthest from the medoid and repeatedly maximized distance to the nearest selected pose. Ties were resolved deterministically in recorded order.

$$d(i,j) = \sqrt{[(\theta_{ij} / 10°)^2 + (\|t_i - t_j\| / 0.1\ m)^2]}. \qquad (2)$$

Equation (2) defines the medoid and geometric policies using training robot transforms only. The scales were fixed analysis choices and were not optimised on validation errors. Subsets were fitted in acquisition order. The main cohort produced 14,832 attempted fits; the ancillary cohort was processed separately. New random streams use seed 20261019 plus a deterministic panel-identifier offset. The original cohort reuses its exact saved subset membership, so processing comparisons remain paired. Every selected pose identifier and failed fit is retained.

3.4 Outcomes and uncertainty

For a held pose v, $Q_v = G_v X C_v$ is its implied fixed-target transform. Translation closure is the Euclidean distance between the translations of Q_v and training reference Z, expressed in millimetres. Rotation closure is the geodesic angle between their rotations, expressed in degrees. Pose-level RMS summarizes each outcome. These errors combine hand-eye estimation, robot kinematics, image-pose estimation and target assumptions; they do not isolate the true error of X.

$$e_t(v) = 1000\ \|t(Q_v) - t(Z)\|; e_r(v) = angle[R(Z)^T R(Q_v)]. \qquad (3)$$

Conditional percentile intervals resample the eight historical validation poses 5,000 times. They describe sensitivity to the empirical held-pose distribution and exclude training uncertainty, systematic measurement errors and robot-population variability. Planned poses are not a random workspace sample, so these are not population accuracy guarantees. Pixel intervals use the same pose-level resampling unit.

Policies are compared within each panel using the farthest-point error divided by the median of 100 random subsets. Ratios below one favour farthest-point selection. Original, added and ancillary cohorts are reported separately. Robot-wise geometric means prevent Robot A receiving extra weight because it has two dates. These are descriptive summaries; panels and random draws are not independent physical trials. No population-level significance tests accompany favourable-panel counts. Numerical failures remain in denominators and are omitted only from finite-estimate summaries.

3.5 Comparators and robustness checks

Full training sets were also evaluated with the Tsai, Horaud, Andreff and Daniilidis options in the same OpenCV version. Nonfinite estimates and invalid rotations were marked unsuccessful. The exact identity-rotation and zero-translation sentinel returned by the Tsai implementation was also treated as failure. Such implementation failures do not establish that the corresponding mathematical method is universally unsuitable.

A secondary fit jointly refined X and Z by minimizing translation and rotation closure with a soft-L1 loss. Translations were scaled by 0.01 m and rotation vectors by one degree; the optimiser used SciPy least_squares with at most 200 evaluations (Virtanen *et al.*, 2020). This objective is a pose-closure refinement, distinct from the joint pixel fit in Section 3.6. Both refinements use training data only and supply alternative objectives, rather than an independent accuracy reference.

Training-pose deletion was repeated for every analysed panel. The initial-cohort record also retains 50 input-order permutations per panel, frame-weighting comparisons and within-pose frame splitting. These checks are identified as initial-cohort sensitivities rather than implied replications on all archives. Separate exact-transform and robot-base-coordinate checks verify the reused implementation conventions. Every panel also checks the vectorized image projection against OpenCV.

For each validation pose, we identified the nearest training pose using Equation (2), reporting its angular and translational separation. This describes the local nature of validation without assuming that distinct pose identifiers guarantee substantial geometric separation. The original sessions were never pooled into a common calibration transform, because camera mounting and robot-model states could differ.

The initial-cohort analysis explored nine geometric-scale combinations. The extension protocol retains the default scales and tests five predetermined combinations spanning 5–20 degrees and 0.05–0.2 m at six and ten poses. This assesses metric dependence without selecting a setting by validation performance. A separate provenance sensitivity excludes archive D, irrespective of its numerical outcomes. Cheirality checks record whether fitted models place observed target corners in front of the camera.

An exploratory objective check used the ten-pose farthest subset and the first ten saved random subsets in every main panel. Each estimate was initialized with its own Park fit and refined on its selected training corners only. The settings were unchanged. Both pipeline ratios used those same ten random draws, keeping subset membership paired while testing objective dependence.

3.6 Raw-image reconstruction and pixel validation

All accepted main and ancillary compressed images were processed locally using the recovered detector, retaining its two-cell black border, tag36h11 codebook, row-major identifiers and canonical corner ordering. Multiscale factors were 1, 1.5, 2, 2.5 and 4; maximum Hamming distance was two, minimum tag area 70 pixels squared, and edge-based corner refinement was used. Iterative OpenCV solvePnP used the recorded, fixed camera matrix and five plumb-bob distortion coefficients. No additional image rectification was applied; the endpoints assess this recorded projection model. Corners, tag identifiers, PnP matrices, per-frame errors, source hashes and parameters were saved.

The primary pose analyses use these newly reconstructed camera-to-target transforms. The same protocol applied to historical exports is retained as a processing-sensitivity comparison. No frame was removed because redetection differed from its historical tag count. A 24-image compatibility check with default one-border Pupil and OpenCV detectors did not reproduce the multi-tag observations (Pupil Labs, n.d.; OpenCV, 2025a). The two-border target definition and recovered settings were therefore specified explicitly; details of the comparator check are retained in the supplement.

For held-image evaluation, each observed target corner was projected through the training estimates of X and Z using its frame-specific robot transform and fixed intrinsics. We averaged squared two-dimensional Euclidean corner errors within each frame, then frames within each pose, then the eight validation poses; the square root is reported in pixels. This gives equal weight to poses despite different frame and corner counts. The camera projection implementation was checked against OpenCV projectPoints to less than 10^{-8} pixels.

A supplementary joint pixel fit optimised the 12 rigid-transform parameters of X and Z on training corners only, initialized by Park. The least-squares objective used the same equal-pose weighting, a linear loss, fixed intrinsics, a maximum of 120 function evaluations and tolerances of 10^{-9}. Held-image errors were then compared with Park and the pose-closure refinement. No held images were used to update parameters or select optimiser settings. Conditional 95% intervals resampled the eight validation poses 5,000 times.

For the two Robot-A dates, we transferred the training-derived, pixel-refined X from one session to the other and kept X fixed. Only destination Z was fitted, using destination training images. The eight direction-by-camera comparisons used destination intrinsics and held images. Uncontrolled mounting history and model changes make this an exploratory stress test, rather than a measurement of mechanical drift.

An additional exploratory transfer analysis paired distinct captures within the same archive, camera role and UTC day, requiring identical recorded camera parameters, target configuration and robot-description hash. Source X remained fixed and only destination Z was fitted on destination training images. Every eligible directed pair was retained. A later online-archive refresh supplied five completed head captures, which were processed separately with the same protocol and excluded from the frozen primary and initial ancillary denominators.

4 Results

4.1 Coverage and local validation geometry

The expanded cohort preserves the distinction between angular excursion and directional coverage (Figure 1). Head, body and wrist sequences occupy different regions of the rotation-span/axis-ratio plane. The archive extension repeats related trajectory families on additional platforms; it does not independently vary robot mechanism, camera role and trajectory design. The axis ratio is therefore used to describe excitation, not to attribute performance to one geometric factor.

The main-cohort validation poses remained local to training geometry: nearest-training separations were 0.23 –3.54 degrees and 1.11–13.66 mm. Distinct pose identifiers consequently do not establish broad workspace separation. These records support a test of local consistency, with limited evidence about larger head rotations, distant wrist configurations or new task workspaces.

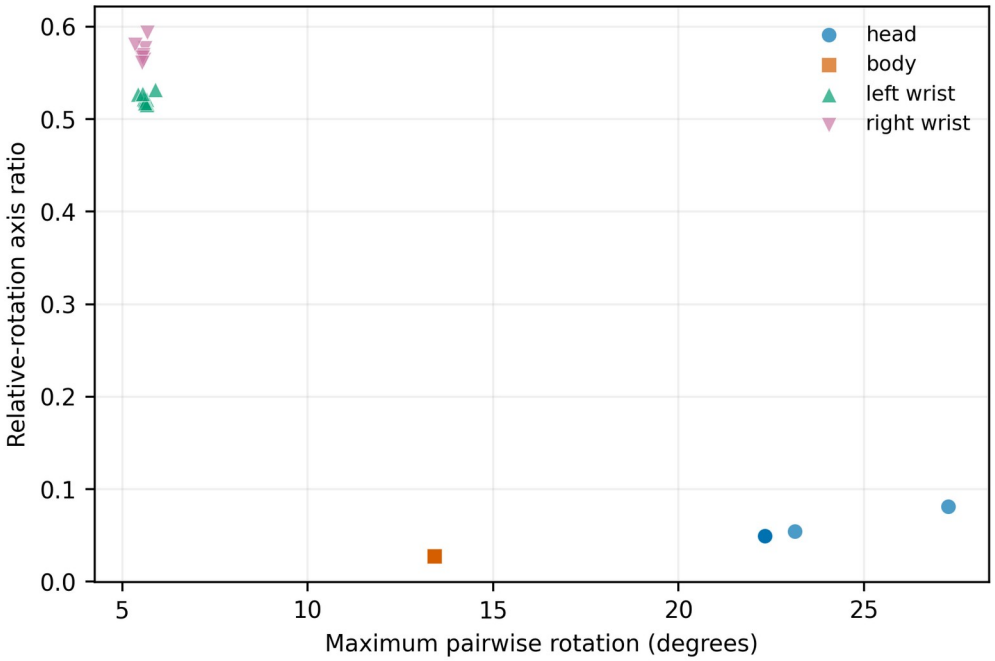


Figure 1. Main-cohort rotation span and relative-rotation axis ratio. Overlapping points reflect related motion designs; the ratio is a geometric descriptor, not a complete identifiability measure. Source: authors' existing records and present calculations.

4.2 Full-set estimates and unsuccessful solvers

The 36 full-set Park estimates were finite. Held-pose translation RMS was 0.256–5.072 mm and rotation RMS was 0.043–0.402 degrees. Table I summarizes the four roles within each session; panel-level values and conditional intervals are in the supplement. Role, kinematic design and viewing geometry covary, so between-role differences do not identify a single cause.

The specified OpenCV implementation produced 27 Tsai and 9 Andreff failures in the main cohort. Table II retains all solver denominators. Finite-estimate medians condition on different successful subsets and cannot be read as a common-panel accuracy ranking. Pose-closure refinement yielded 0.202–2.507 mm held-pose translation RMS; the held-image endpoint provides a separate check on that objective change.

OpenCV 4.11.0 discards Tsai motion pairs when either scaled minimal-quaternion vector has norm outside 0.3–1.7, and returns without an estimate if fewer than two pairs remain (OpenCV, 2025b). The lower threshold corresponds to about 17.3 degrees of relative rotation. The retained-pair audit distinguishes this implementation rule from a general mathematical failure. Body and wrist trajectories below the threshold cannot provide retained pairs under that rule.

[Insert Table I about here]

[Insert Table II about here]

4.3 Pose budget and selection policy

At ten poses, farthest-point selection gave lower translation closure than the random median in 9/12 original panels, with a median ratio of 0.833. The added archives gave 17/24 favourable panels and a median ratio of 0.879 (Figure 2; Table III). Individual added-cohort ratios ranged from 0.475 to 1.779. Four of six archive summaries were below one, with an archive-balanced ratio of 0.883.

Held-image error favoured farthest-point selection in 17/24 added panels, with a median ratio of 0.879 and an archive-balanced ratio of 0.919. Across all 36 main panels, median ratios were 0.854 for translation closure and 0.912 for held-image error (Table III). A favourable closure comparison need not imply a favourable image comparison.

The relationship changed with pose budget (Figure 3). The recorded design provides more selection freedom at six poses than near the full 25- or 30-pose set. Error did not decrease monotonically in every panel, and a subset could outperform its full-set fit. Table III and the source tables retain the denominators and ratios for each cohort, rather than pooling random draws into a hardware sample.

The main cohort contained 49/14,832 numerically unsuccessful subset fits. The failure curves in Figure 3 distinguish numerical failures by policy. One additional finite ten-pose chronological estimate predicted negative target depth and remains separately flagged; it is not counted as a successful physical calibration. The 23 ancillary panels gave 20/23 panels, median ratio 0.885 by closure and 19/23 panels, median ratio 0.885 by image error at ten poses. Those captures involve additional dates, software-development sequences and incomplete role coverage; they test sensitivity without expanding the matched four-role primary cohort.

The matched objective check substantially attenuated the image-based selection contrast (Figure 4b; Table S7). Across the same 36 panels and eleven subsets per panel, the median ratio was 0.911 under Park and 0.993 after pixel refinement. The refined archive-balanced ratio was 1.001; 22/36 panels were below one. All 396 refinements converged and predicted positive target depth on the held images. This exploratory check used ten random subsets, rather than the primary baseline of 100.

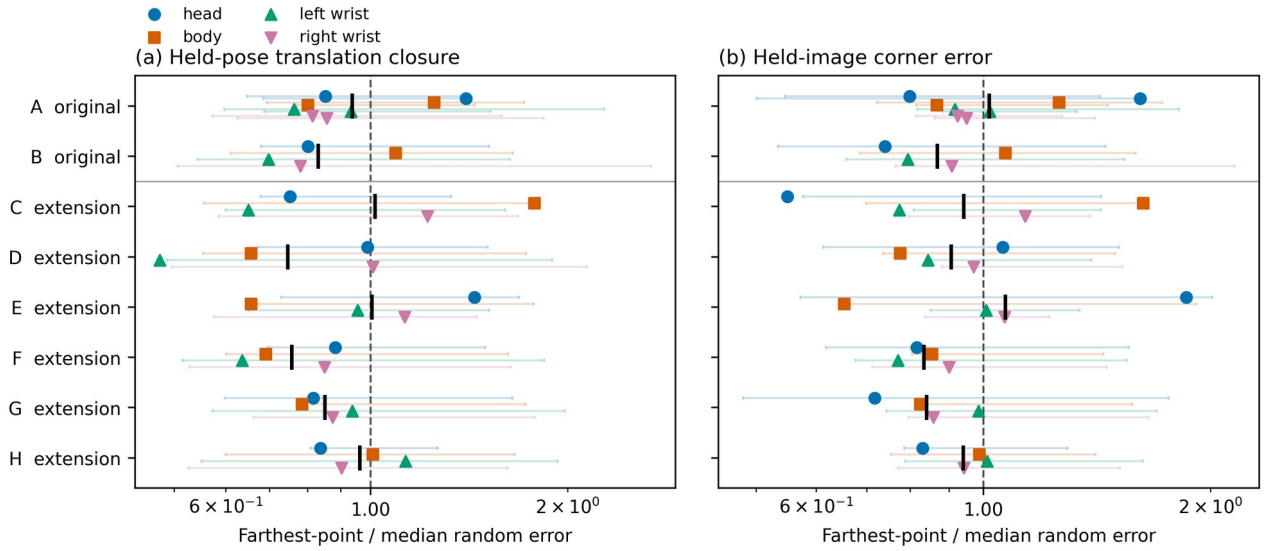


Figure 2. Ten-pose farthest-point error divided by the median of 100 random-subset errors, separately for (a) translation closure and (b) equal-pose held-image error. Points are camera-session panels; horizontal ranges show the central 95% of finite random-subset errors, normalized by their panel median. These are algorithmic ranges, not confidence intervals. Black ticks are within-archive geometric means. A and B form the original cohort, C–H the added cohort. The dashed reference is one. All fits use training data only; random draws are algorithmic replicates. Source: present calculations.

[Insert Table III about here]

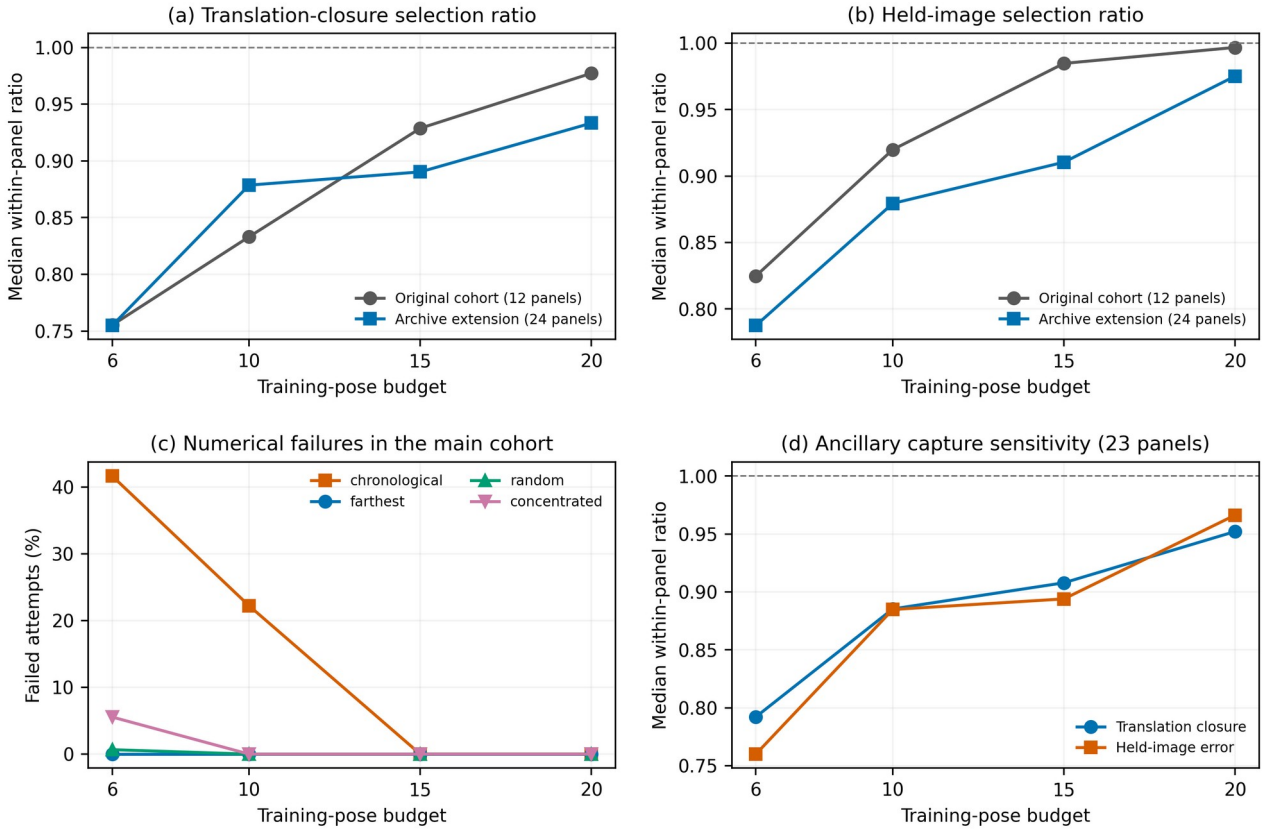


Figure 3. Median within-panel farthest/random ratios by budget for (a) closure and (b) held-image error, with original and added cohorts separated. (c) Numerical-failure percentages use 36 fits per deterministic policy and 3,600 random fits per budget. (d) Ancillary-cohort ratios are reported separately. Each policy fits the same historical training pool and validation split. Source: present calculations.

4.4 Sensitivity and provenance

Omitting one training pose changed X by up to 8.554 mm in translation and 0.463 degrees in rotation across main panels. These are finite-design sensitivities, not uncertainty bounds on true mounting error. The initial-cohort order-permutation and frame-split results remain in the research archive and are not presented as new hardware replicates.

The five prespecified scale combinations yielded ten-pose median closure ratios of 0.847–0.883 in the main cohort. Excluding archive D gave 23/32 favourable panels by closure and 22/32 by image error, with median ratios of 0.854 and 0.912. This sensitivity retains D's incomplete historical identity evidence without treating its storage hostname as definitive acquisition identity.

4.5 Raw-image outcomes and cross-session transfer

All 6,383 accepted main-cohort timestamps were recovered and redetected. Tag counts ranged from 22 to 35; 5,966 frames reproduced the stored tag-identifier set. Median reconstructed-to-exported PnP differences were 0.129 mm and 0.050 degrees. Frame-specific raw TF comparisons agreed to numerical precision. CameraInfo fields and model hashes matched all 72 main recording contexts; the two ancillary hash exceptions are documented separately.

Full-set Park held-image RMS was 0.278–2.732 pixels. Training-only pixel refinement gave 0.247–0.616 pixels, with convergence recorded in 36/36 panels (Figure 4). Pose-closure refinement is shown on the same image endpoint. These are errors from one shared kinematic projection per panel, distinct from PnP reprojection after independently fitting every image. The initial archived-parameter comparator remains in the research archive; similar image errors can coexist with different X translations.

The original-cohort date-transfer diagnostic was retained unchanged. Six head, body and left-wrist comparisons had 0.99–1.14 times the destination-specific held-image error. The two right-wrist directions gave 89.03 and 97.84 times, coinciding with a 179.75-degree difference between session-specific X rotations. Every transfer fixed source X and fitted only destination Z on destination training images. Mounting history, image-coordinate configuration and kinematic changes remain uncontrolled, so the result does not establish mechanical drift. The archive extension supplies additional role-level consistency evidence, not controlled longitudinal replication of that discrepancy.

The matched same-day analysis comprised 52 directed transfers across three archives (Table S5). The median transferred/destination-specific held-image ratio was 1.005, with a range of 0.922–1.199. Matching recorded configuration narrows the comparison but does not establish unchanged physical mounting. Pairs share captures and are not independent trials. The five late-available head sequences had larger full-set Park errors (8.818–9.957 mm closure and 5.487–6.047 pixels); their pixel-refined errors were 0.617–0.718 pixels (Table S6). These single-role captures remain outside the frozen cohorts but limit extrapolation from the frozen-cohort residual range.

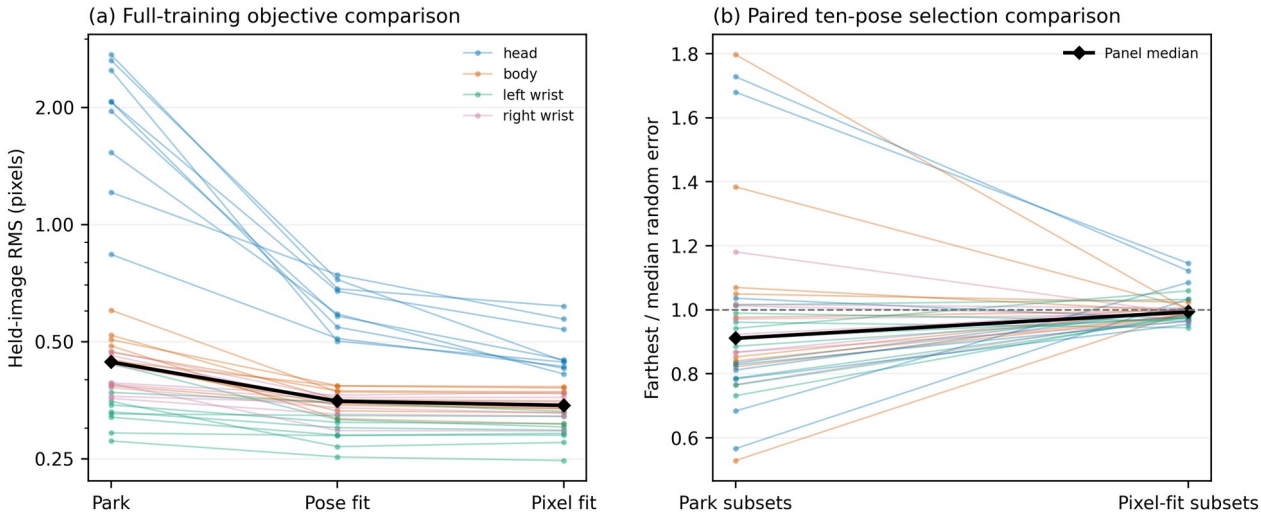


Figure 4. (a) Full-training objective comparison on held images. (b) Ten-pose farthest/random image-error ratios before and after training-only pixel refinement, using the same farthest subset and first ten saved random subsets per panel. The random denominator here differs from the 100-draw primary comparison in Figure 2. Coloured lines are main-cohort panels; black diamonds are panel medians. All fits use training data only and fixed recorded intrinsics. Source: present calculations.

5 Discussion

The archive extension supports a conditional small-budget advantage for farthest-point selection under Park fitting. The paired subset check shows that this contrast approaches unity after training-only pixel refinement. Geometry therefore remains relevant to initialization and solvability, but its apparent benefit depends on the fitted objective and validation endpoint. The evidence does not establish a universally preferable selection policy.

This evaluation complements configuration design by Wu *et al.* (2015), uncertainty-based active selection by Yang *et al.* (2023), and reinforcement-learning pose selection by Li *et al.* (2026). The present policy cannot generate new poses or model measurement uncertainty. Its practical contribution is a comparison over physical archives with matched budgets, raw-image traceability, failed fits and explicit extension/ancillary cohorts. Related trajectory templates across the platforms still restrict generalization.

Closure and image residuals test related models. Closure compares implied target poses with a training-only reference; image error asks whether a shared fitted kinematic projection explains held corners. Changing the objective can redistribute kinematic, intrinsic and target errors, as motivated by multi-metric calibration work (Tabb and Ahmad Yousef, 2017). Joint kinematic/extrinsic methods such as Vieira *et al.* (2026) address additional parameters that are fixed here. Neither of our endpoints isolates true camera-to-link error.

Data lineage is part of the experimental design. Counting copied reference exports or rerun analyses as new hardware trials would have inflated the cohort. Matching raw recordings, timestamp overlap and image payloads instead links software outputs to physical captures. The two-cell target border also shows why a family name alone is insufficient to reproduce corner extraction. The retained implementation and decoded corners make compatible reprocessing inspectable without claiming exact recording-date software identity.

The date-transfer diagnostic and matched same-day analysis test conditions beyond a local holdout. Identical recorded intrinsics and model text reduce some configuration differences in the latter, but cannot establish unchanged physical mounting or image coordinates. Shared captures also make directed-pair outcomes dependent. These checks motivate a configuration-continuity assessment rather than a general claim of mechanical drift or temporal instability.

Industrial evaluation should preserve the target definition and image-processing settings, retain failures at the intended budget, and test omitted poses with stated endpoints. External reference measurements or task outcomes are then needed to connect consistency to positioning requirements (Bai *et al.*, 2020; Wang *et al.*, 2025). The present ratios are empirical comparisons, rather than acceptance thresholds.

These platforms carry multiple cameras, but the recordings were separate role sequences on different support links. Shared-target and camera-to-camera constraints of the kind used by Allegro *et al.* (2024) were not tested. Synchronized common-target observations would be needed to extend the findings to inter-camera registration.

6 Limitations and next validation steps

The cohort is conditioned on existing exports, retained recordings and historical capture-window screening. Traversing additional storage did not make inaccessible devices observable. Sessions failing before an export, unlinked recordings and the four reference groups without raw bags remain outside numerical comparisons. We retained failed fits within eligible panels, but this does not remove acquisition, availability or developer-selection bias.

There was no external metrology, independently measured target dimension, controlled mounting-repeatability experiment or application-task evaluation. Raw TF reconstruction verifies the saved transform chain, not the physical accuracy of the robot kinematic model. Fixed intrinsics, upstream image rectification, target flatness, timing effects and corner-localization bias may affect every fitted model. Exposure settings and the pre-recording imaging pipeline were not independently reconstructed. Image encoding also varied between archives, so between-archive differences cannot be assigned solely to robot or camera performance.

Eight robot archives and nine complete sessions provide wider coverage than the initial study, but they share a platform family and related trajectories. Camera roles, dates and repeated frames remain dependent. Archive D lacks a retained session robot identifier, motivating its exclusion sensitivity. The historical holdouts are local and had been available to the original workflow. Pose bootstrap intervals do not cover systematic errors or hardware-population variability.

The geometric metric, objective scales and optimiser settings are fixed choices, not a comprehensive comparison with active or uncertainty-weighted designs. The added archives test a frozen retrospective protocol, not prospective policy deployment. A broader untouched validation session with independent metrology would directly test whether the observed consistency patterns transfer to positioning performance.

The local replication contains accepted images, corners, extracted contexts and numerical reconstruction, but not full raw-bag replicas or recording-date environment lockfiles. Some archives contain development reruns and regression exports; deduplication and separate ancillary reporting reduce, but cannot remove, their selection history. Public data access and a persistent repository identifier require an author-finalized sharing arrangement.

7 Conclusions

A frozen protocol applied across eight robot archives found a Park-based small-budget advantage for farthest-point selection. In the 24 added panels, ten-pose comparisons favoured it in 17 panels by both translation closure and held-image error, with median ratios of 0.879. The paired pixel-refinement check shifted the median image-error ratio to 0.993 and its archive-balanced ratio to 1.001. The apparent selection benefit therefore depended on the training objective within the tested local validation design.

The comparison is grounded in 6,383 accepted main-cohort images, with ancillary and late-capture checks reporting broader conditions. These results assess conditional calibration consistency. Absolute mounting accuracy, broad workspace transfer and inter-camera registration require independent measurements.

Data and code availability

The measurement-level data and analysis code are packaged for local replication, with direct device and network identifiers removed. Complete image evidence, recording locators and provenance records remain in the private study archive. The materials are not yet publicly deposited; a permanent repository or controlled-access link is pending author finalization.

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